# SoS'26 Plan of Action

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# 1 Introduction

The goal of this SoS project is to build strong foundations in Data Structures and Algorithms using C++

The primary focus of this project will be:

- Understanding core programming concepts in C++
- Learning fundamental data structures and algorithms
- Developing problem-solving skills
- Practicing problems regularly on LeetCode for understanding concepts

# 2 Plan of Execution

I plan to do the week's work spread across all the days of the week, with a focus on consistency and regular practice. The schedule will be as follows: I will learn the topics from YouTube lectures and slides of Ashutosh Gupta and Rohit Gurjar and then practice problems on LeetCode. I will also refer to CLRS for deeper understanding.

# 3 Weekly Division of Topics

## Week 1: C++ Basics and Elementary Algorithms

- Variables and Data Types
- Loops and Conditional Statements
- Functions
- Arrays and Arrays in Functions
- Selection Sort
- Bubble Sort
- Insertion Sort
- Pointers
- 2D Arrays
- Strings and Functions on Strings
- Passing by Value and Passing by Reference
- Time Complexity Analysis
- Chapters 2 and 3 of CLRS

## Week 2: Recursion and Searching/Sorting Techniques

- Recursion
- Binary Search
- Merge Sort
- Quick Sort
- Structures

## Week 3: Stacks, Queues and STL

- Stacks and Queues
- STL for Stack and Queue
- Deque and Circular Queue and Simple Queue
- Vectors

## Week 4:clrs and techniques

- CLRS Chapters 4, 7 and 11
- Two Pointer Technique
- Sliding Window Technique
- Sets and Unordered Sets (Basic Understanding)
- Classes and Constructors (Basic Understanding)

********** Midterm Submission **********

## Week 5:linked lists and priority queues

- Singly Linked List
- Doubly Linked List
- Fast and Slow Pointer Technique
- Cycle Detection in Linked List
- Prefix Sum Arrays
- Difference Arrays
- Priority Queues

### Week 6:priority queues, Heaps and Priority Queues

- Basic Hashing Concepts
- Hash Maps
- Heaps
- Heap Sort
- CLRS Chapter 6

### Week 7: Trees and Binary Search Trees

- Binary Trees
- Binary Search Trees
- Recursion on Trees
- Recursion Tree Method
- CLRS Chapter 12

### Week 8: graph algorithms

- Introduction to Graphs
- Breadth First Search (BFS)
- Depth First Search (DFS)
- Topological Sorting
- Graph Representation
- CLRS Chapter 22

********** Endterm Submission **********

## 4 Resources

- Introduction to Algorithms (CLRS) by Cormen, Leiserson, Rivest and Stein
- Prof. Ashutosh Gupta's Lecture Slides
- Striver's YouTube Lectures for DSA for reference
- Rohith V G's CS101 Slides for C++ Basics
- LeetCode for Problem Solving Practice